Basic Principles of Medicine 2
Module: Digestion and Metabolism
Lecture 20

Functions and Regulation of the GI Tract

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Note: The lecture slides that are available for download and print-out contain complementary slides that are not presented during the lecture.

These slides have a blue background and contain useful information about the material discussed during the lecture.

Recommended Reading

Rhoades, R. & Bell, D. (2023); *Medical Physiology: Principles for Clinical Medicine, 6*th Edition; Wolters Kluwer. Chapter 25

[Gastrointestinal Motility | Medical Physiology: Principles for Clinical Medicine, 6e | Premium Basic Sciences | Health Library]

Practice Question Sources	Location
Clinical Key - Guyton & Hall Physiology Review, Unit XII, 195-214	https://www.clinicalkey.com/student/content/book/3-s2.0-B9780323639996000120
Practice Questions	BPM501 Sakai Site
Other Resources – BRS, Pretest	University Library

How will the GI Physiology be covered?

Activity	Activity Title
Lecture 20	Functions and Regulation of the GI Tract
Lecture 22	Enteric Motility
Lecture 24	GI Secretion-I
Lecture 26	GI Secretion-II
Lecture 39	Flipped Class: Clinical Cases (Biochemistry and Physiology)

Lecture Objectives

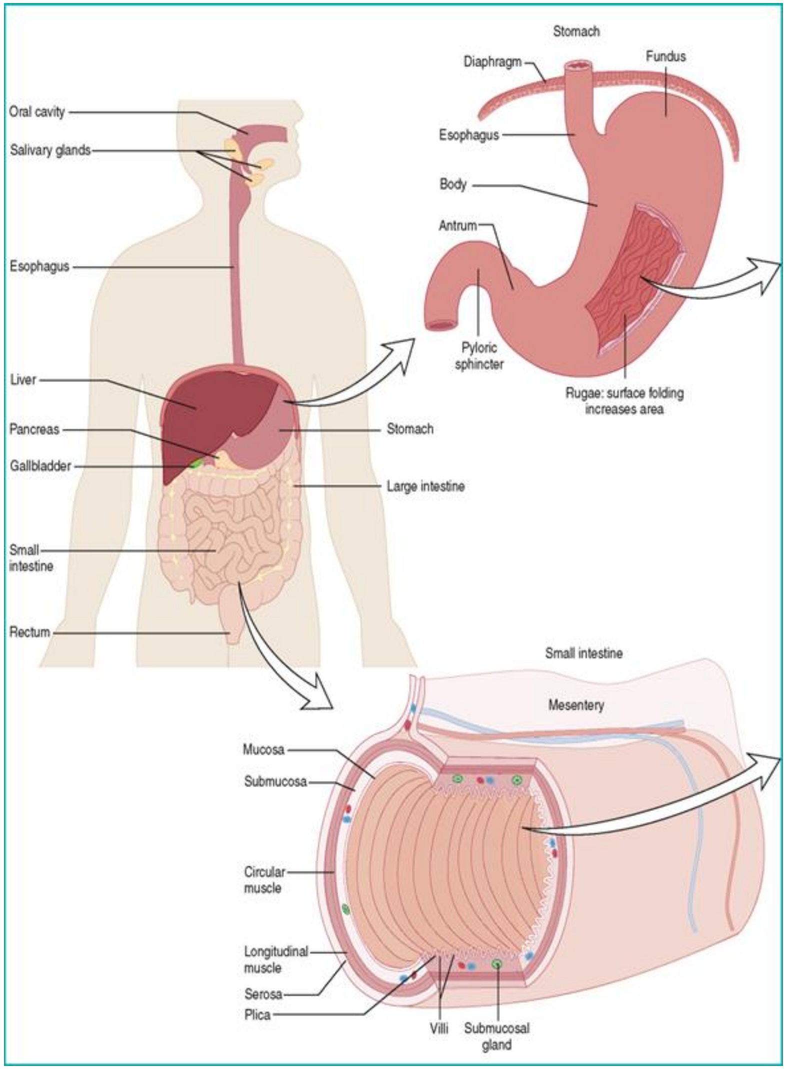
Objective Code	Objective Text
SOM.MK.II.BPM2.3.DM.1.PHYS.1095	Differentiate the processes of ingestion, digestion, absorption, secretion and excretion; include the location in the GI tract where each process occurs
SOM.MK.II.BPM2.3.DM.1.PHYS.1096	Outline the major anatomical characteristics of the enteric nervous system and the key cell types in enteric ganglia (sensory nerves, interneurons, and motor neurons). Given either a cross section of bowel wall, identify the anatomical positions of the myenteric and submucosal plexi, and list which muscle layers they control.
SOM.MK.II.BPM2.3.DM.1.PHYS.1097	Explain how afferent and efferent extrinsic nerves (sympathetic and parasympathetic) interact with the enteric nervous system to regulate functions of the GI tract.
SOM.MK.II.BPM2.3.DM.1.PHYS.1098	Compare and contrast the regulation of gut function by nerves, hormones, and paracrine regulators.
SOM.MK.II.BPM2.3.DM.1.PHYS.1099	List the major excitatory and inhibitory motor neurotransmitters and digestive hormones in the GI tract and their key regulatory functions.
SOM.MK.II.BPM2.3.DM.1.PHYS.1100	Analyze how GI cells integrate regulatory inputs and explain how the ultimate behavior of GI tissues results from summation of inputs from multiple regulatory pathways.

Lecture Outline

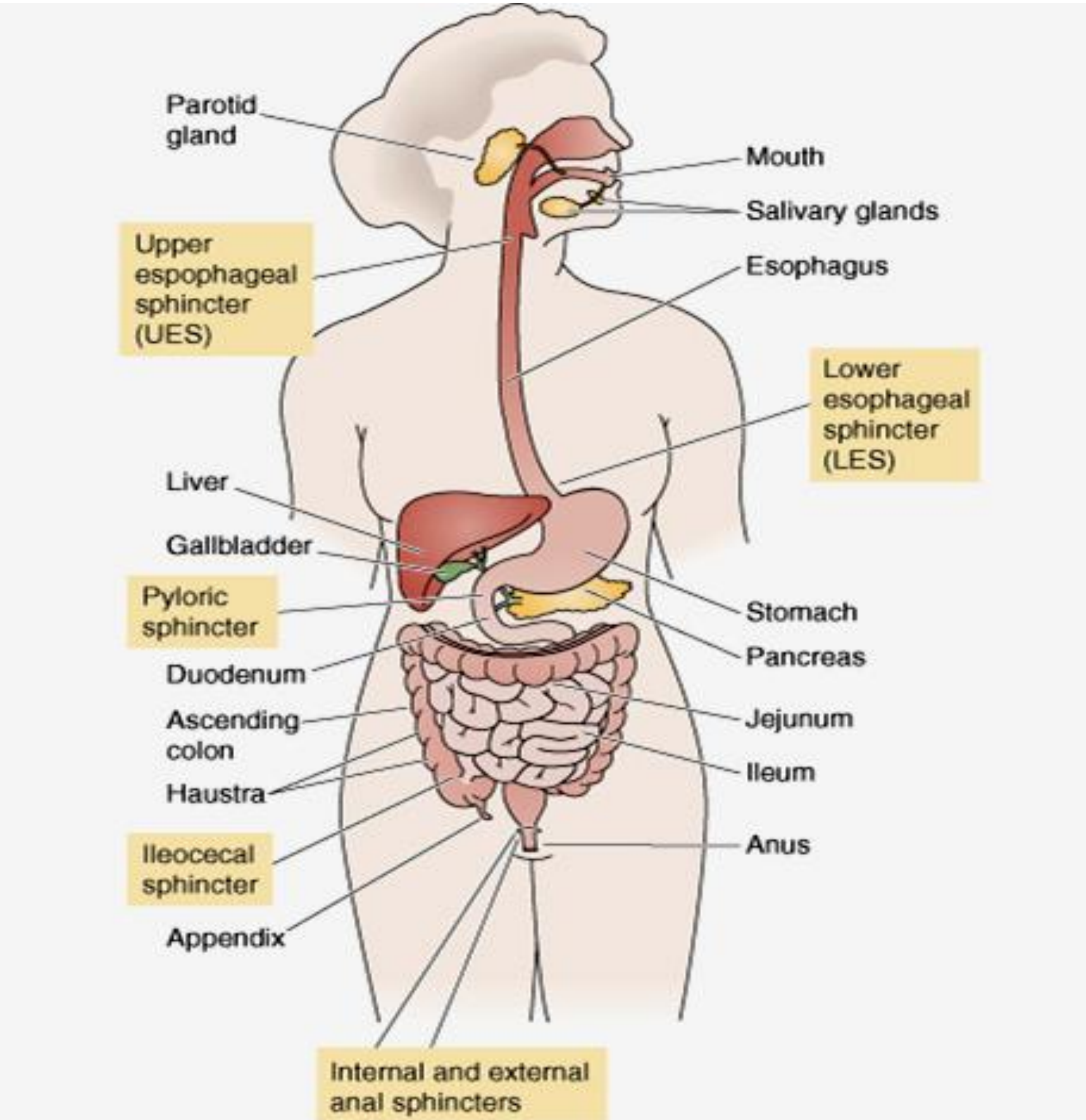
- Overview of GIT structure and function
  - macroscopic and microscopic levels
  - links between the structure and regulatory systems
  - enteric and autonomic nervous system
- GIT regulators
  - Endocrine
  - Neurocrine
  - Paracrine
  - Integration of regulatory pathways
- Stages of meal processing

Part 1

GIT Structure and Function Overview



Gastro-intestinal tract (GIT) macro structure



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GI Tract:

Esophagus

Stomach

Small Intestine

Large intestine

Anus

Accessory Organs and Glands

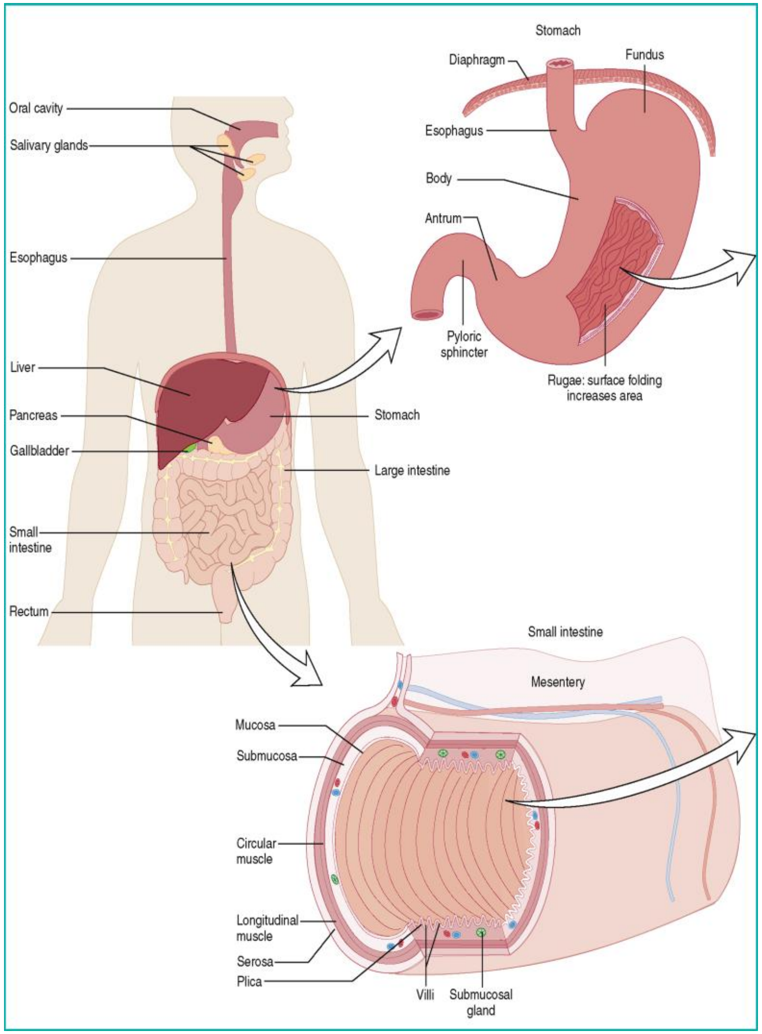
Salivary glands

Liver

Gall bladder

Pancreas

GIT functions overview



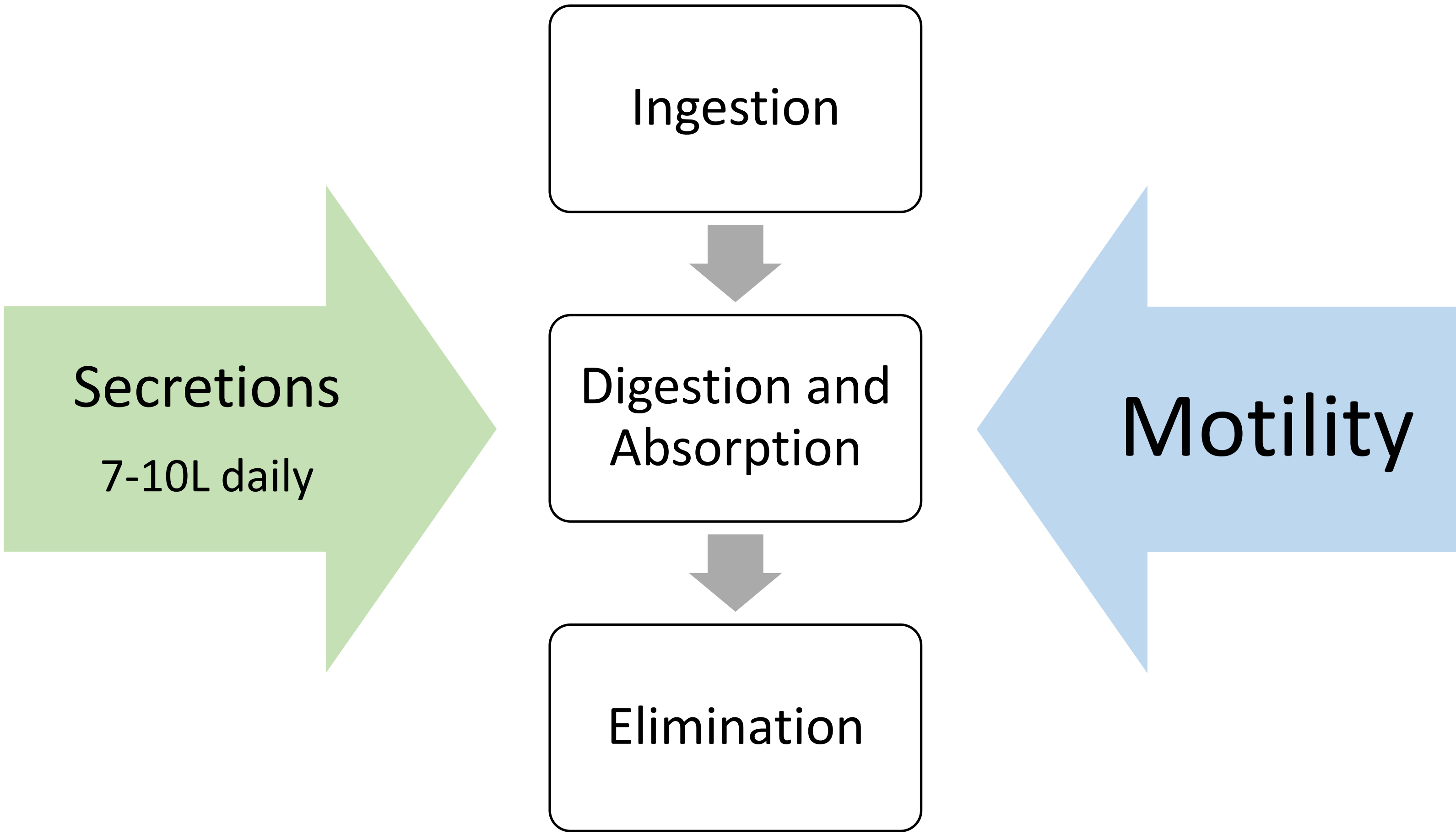
- **Ingestion** - the taking of food, drugs, liquids, or other substances into the body by mouth
- **Digestion** - the breakdown of food into smaller components that can be absorbed into the bloodstream
- **Absorption** - the movement of nutrients, water and electrolytes from the lumen of the GIT into the bloodstream
- **Secretion** - production and release of a substance by a gland or cell
- **Excretion** - the process of eliminating or expelling waste matter.

The overall function of the G.I. system is to absorb nutrients, electrolytes and water and eliminate waste products.
Mechano- and chemo receptors respond to: Stretch, osmolarity and pH; Presence of substrate and end products of digestion. They initiate reflexes that activate or inhibit certain digestive glands Physiological processes to achieve these include motility, secretion of digestive juices, digestion of the food, absorption of the digestive products and fluids, the circulatory system to carry the absorbed substances away and the control of all these functions by the nervous and hormonal systems.

Most food is taken in as solids and as macromolecules that cannot be readily transported across the intestinal epithelial cells. Digestion which involves physical and chemical alterations of the food has to first take place and this is facilitated by secretions from the G.I. tract and associated organs.

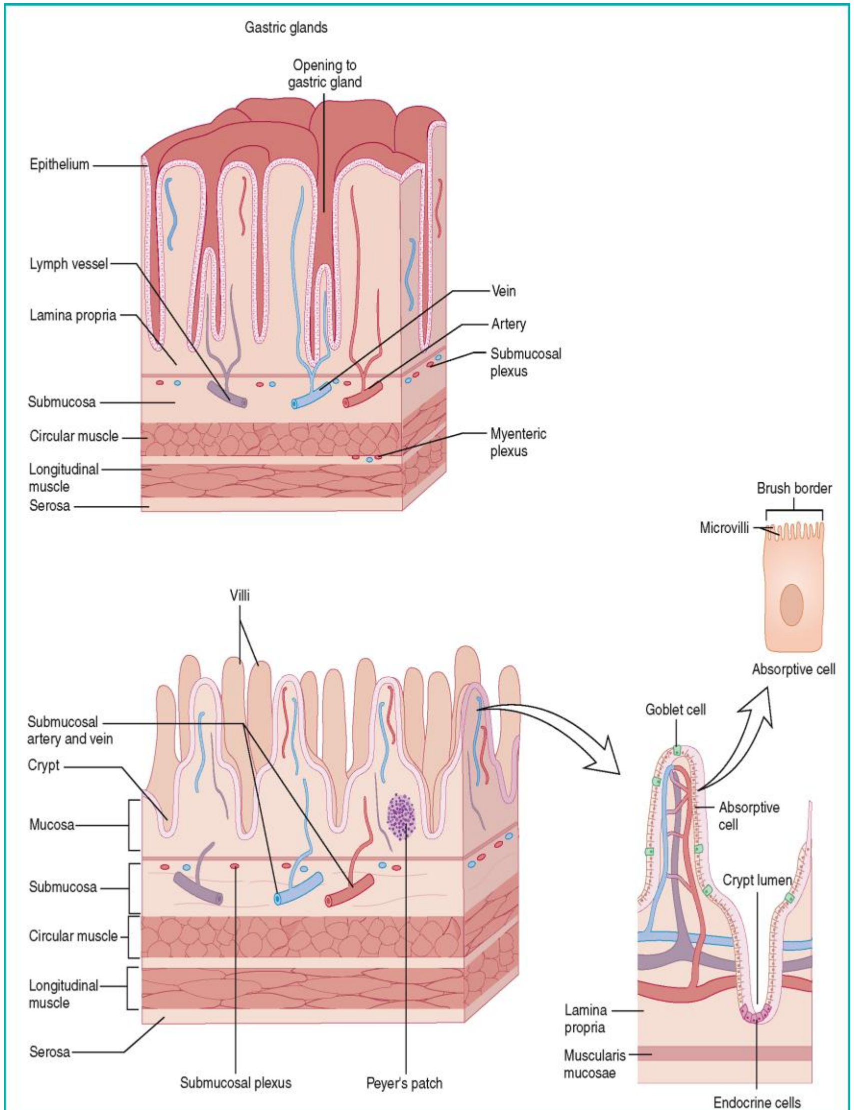
The absorbed nutrients are then metabolized - through a series of chemical processes - making it possible for cells to continue living. Waste substances from the ingested food are stored and excreted in the G.I. tract. Also excreted in the G.I. tract are products from the liver such as bile acids formed from cholesterol, bile pigments formed from hemoglobin degradation and drug metabolites.

Integrated Control of GI Functions



Digestion and absorption of food is coordinated with secretions from glands and motility of the GIT

Microscopic overview



- **Epithelial cells** - are specialized for various secretions and absorption of nutrients

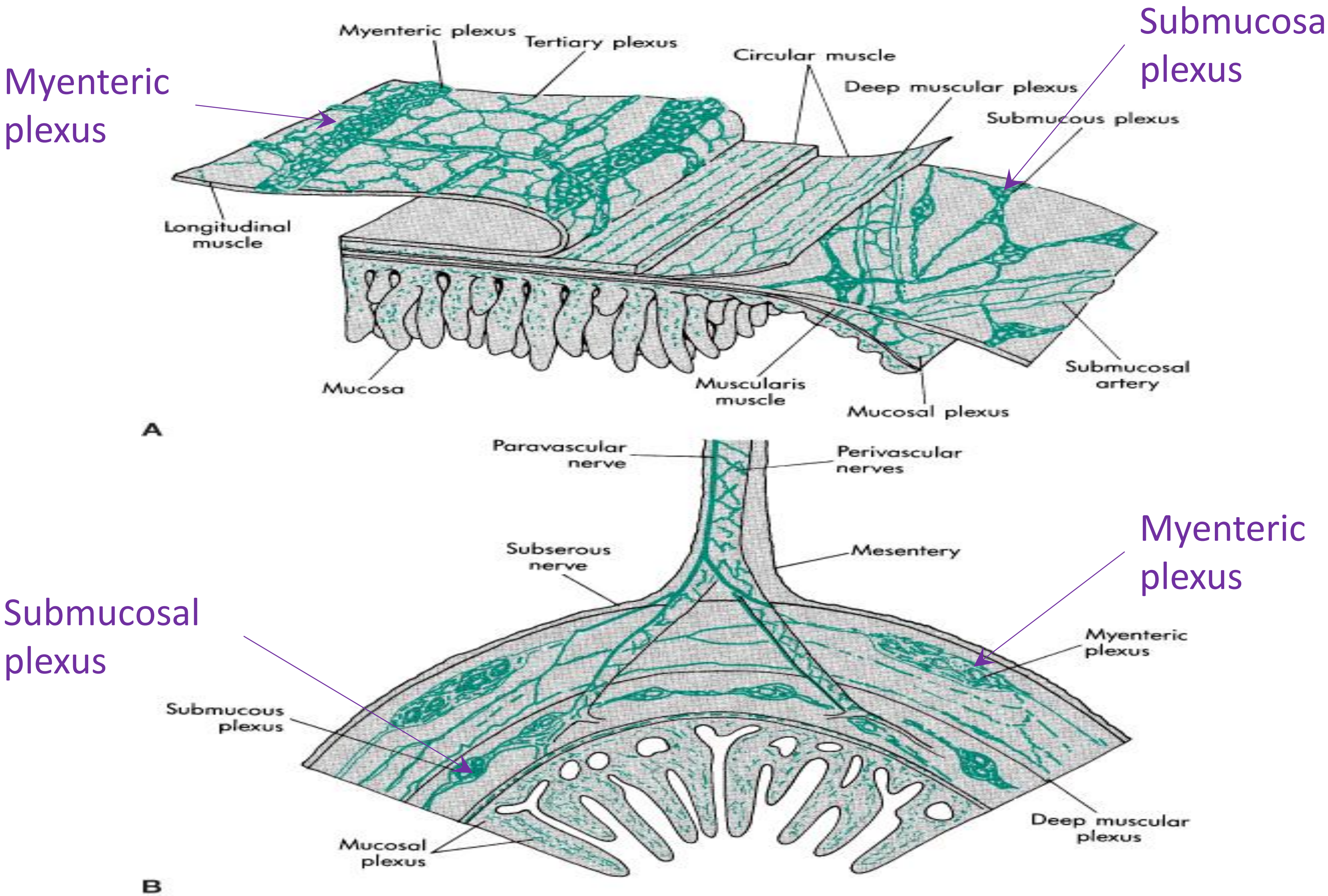
- **Muscularis mucosa** – contractions change the surface areas for secretion or absorption

- **Circular muscle** – contractions decrease diameter of the lumen

- **Longitudinal muscle** – contraction shortens the segment of the GIT

- **Submucosal plexus (Meissner) and Myenteric plexus** – integrate the motility, secretory and endocrine functions of the GIT.

The Enteric Nervous System

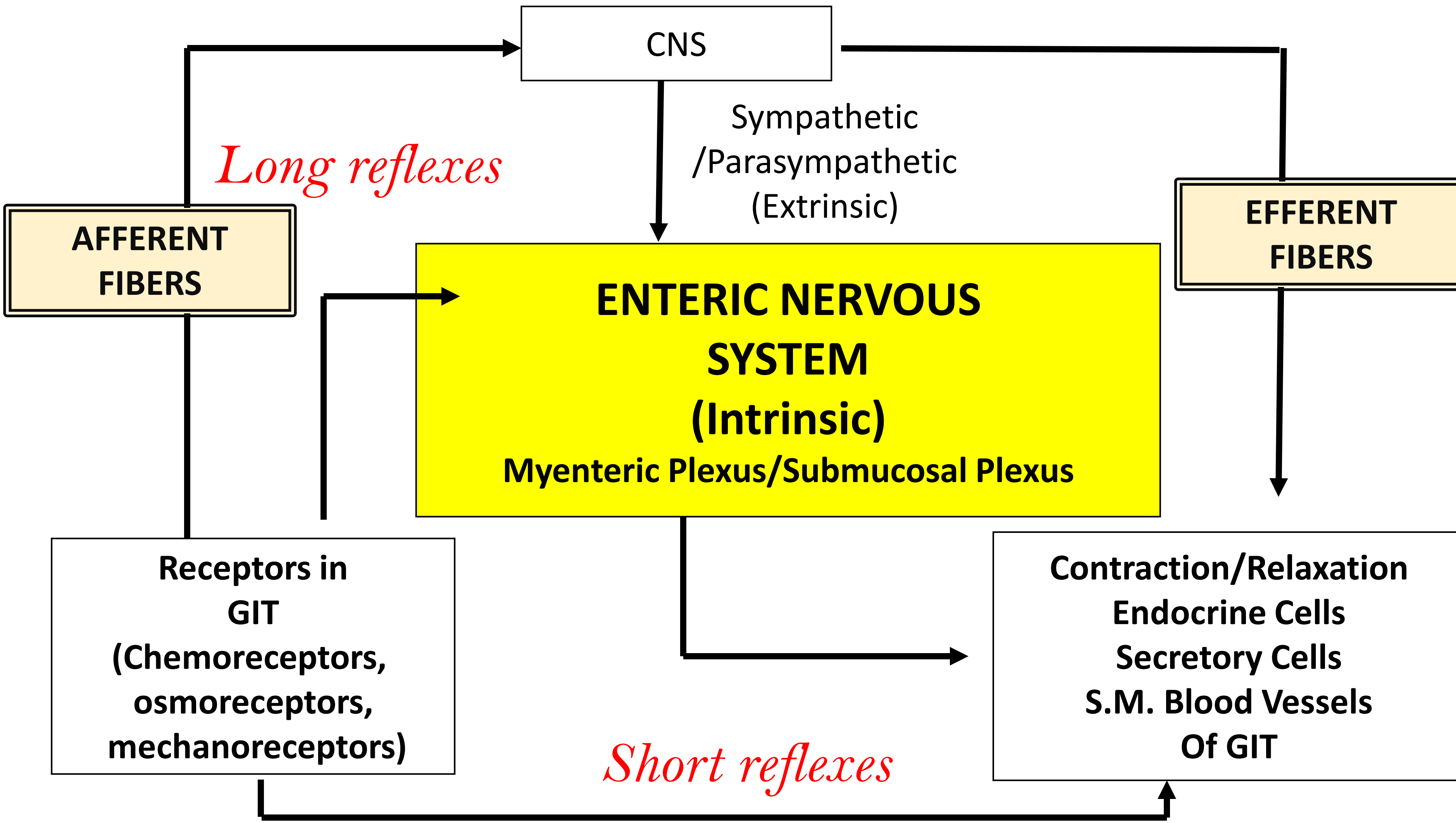


The enteric nervous system (ENS) is a “mini-brain” with independent information processing capability, and also a “program library” that initiates patterns of GI activity (e.g., inter-digestive, digestive, emetic activity, etc).

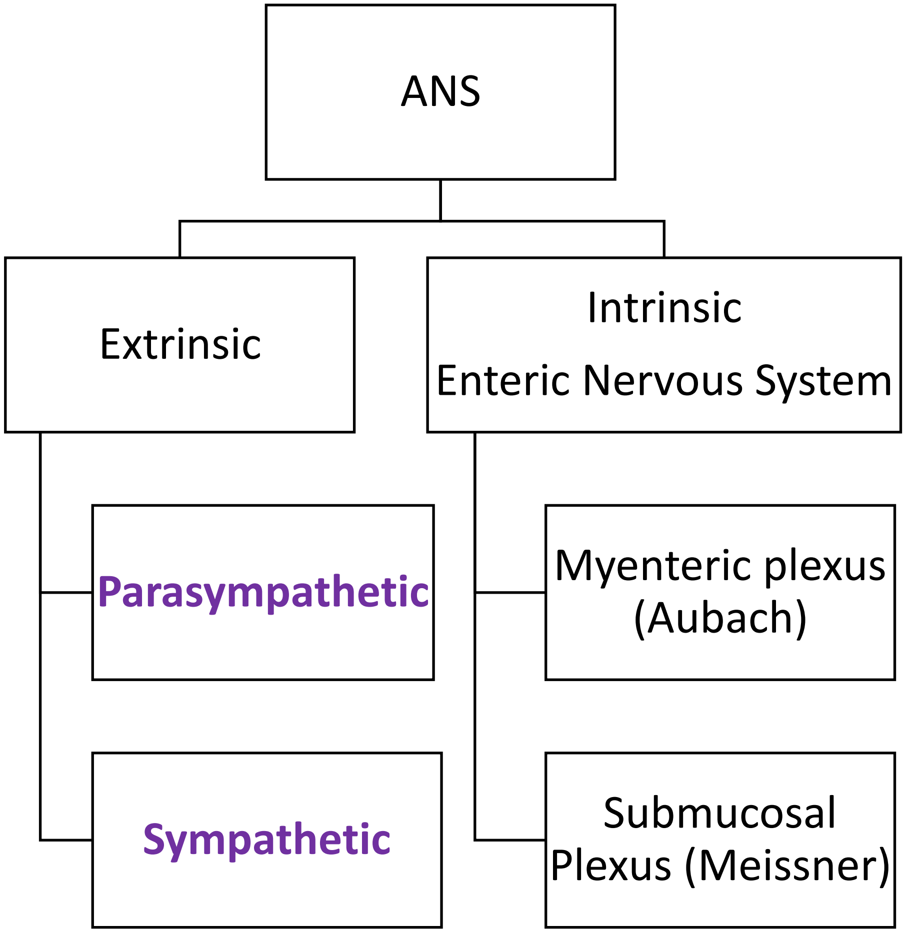
The ENS is arranged anatomically into nerve networks in the **myenteric and submucosal plexuses**. The system contains around 100 million neurons (sensory, inter- and motor neurons). Ganglion cells in the ENS resemble those in the CNS to form sophisticated integrated neural networks. Given its size, the ENS is often viewed as a “displaced” part of the CNS, concerned solely with regulating GI function.

The ENS can function autonomously without extrinsic connections. The **myenteric (Auerbach’s) plexus** is primarily involved with **control of motility** and innervates the longitudinal and circular smooth muscle layers. The *submucosal (Meissener’s)* plexus mainly innervates the glandular epithelium, intestinal endocrine cells and submucosal blood vessels to control intestinal secretion.

Enteric Nervous System Overview



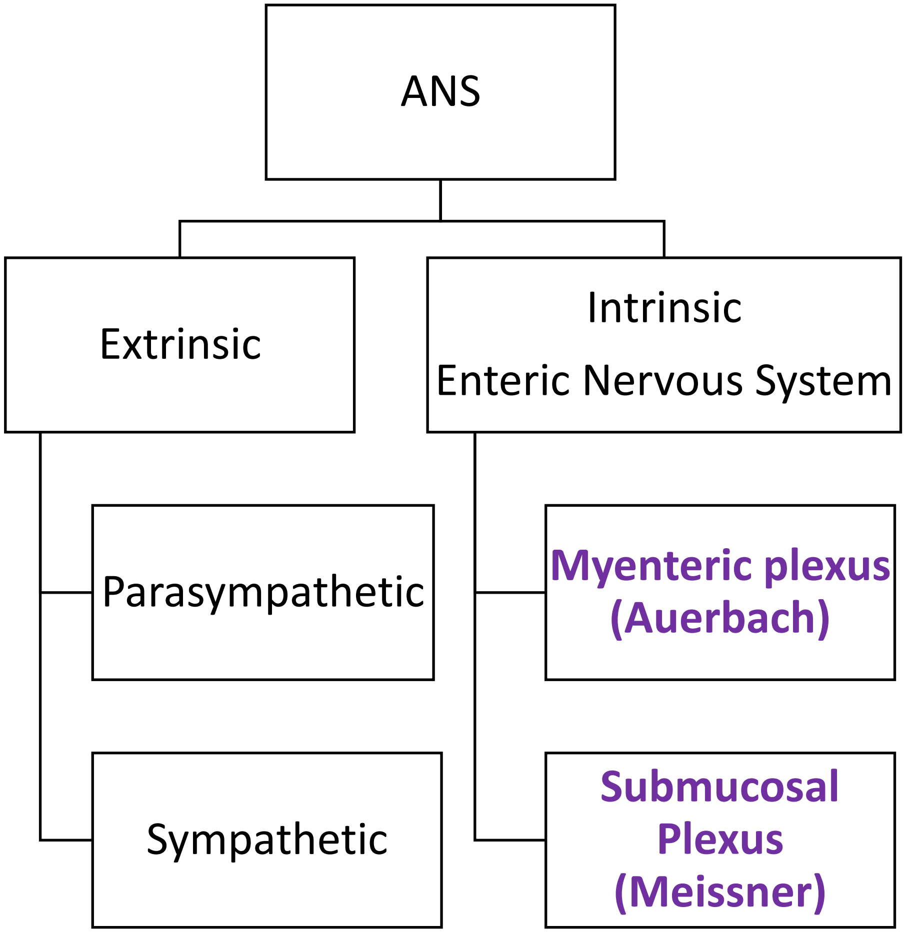
Extrinsic Innervation of the GIT



- **Parasympathetic**
  - Excitatory
  - Vagus (Upper GIT) and pelvic nerves (Lower GIT)
  - Preganglionics synapse on plexi
  - Postganglionics synapse on smooth muscle, secretory and endocrine cells

- **Sympathetic**
  - Inhibitory
  - Thoracolumbar fibers (T5-L2)
  - Preganglionics (cholinergic) synapse in the prevertebral ganglia
  - Post ganglionic (adrenergic) synapse on plexi
  - Plexi relay info to target cells

Intrinsic Innervation of the GIT

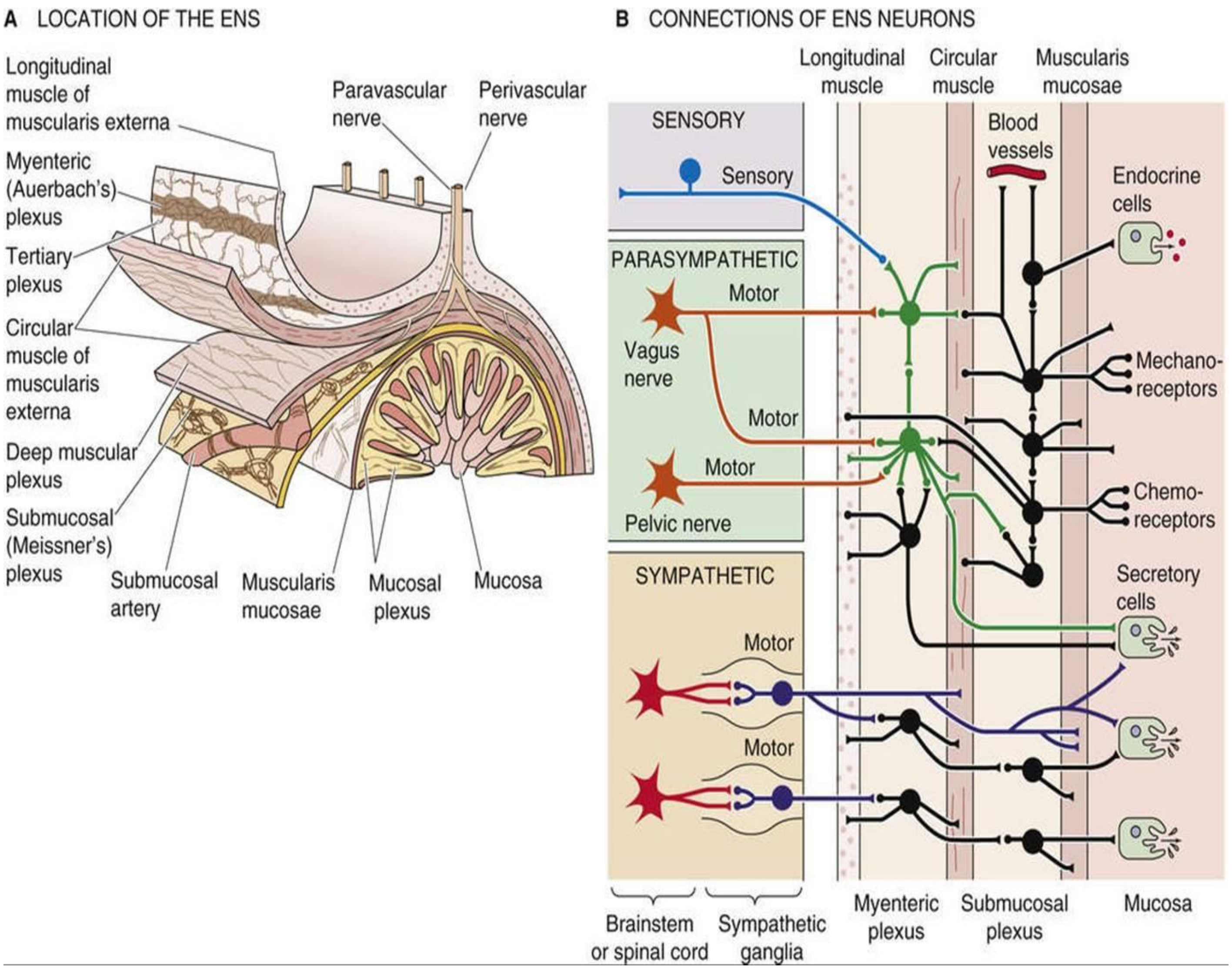


- Extrinsic input (from ANS), but ENS can function in the absence of extrinsic input
- uses local reflexes

- **Myenteric plexus**
  - control of enteric motility

- **Submucosal**
  - regulates secretions and blood flow
  - receives sensory information from chemo- and mechanoreceptors

Neural Control of Motility and Secretions



https://basicmedicalkey.com/physiology-of-the-digestive-system/

Parasympathetic: excitatory

Sympathetic: inhibitory

Chemoreceptors/ mechanoreceptors: receive stimuli stretch/distension, composition of fluid

Endocrine cells: regulatory peptides

Extrinisic Neural Control

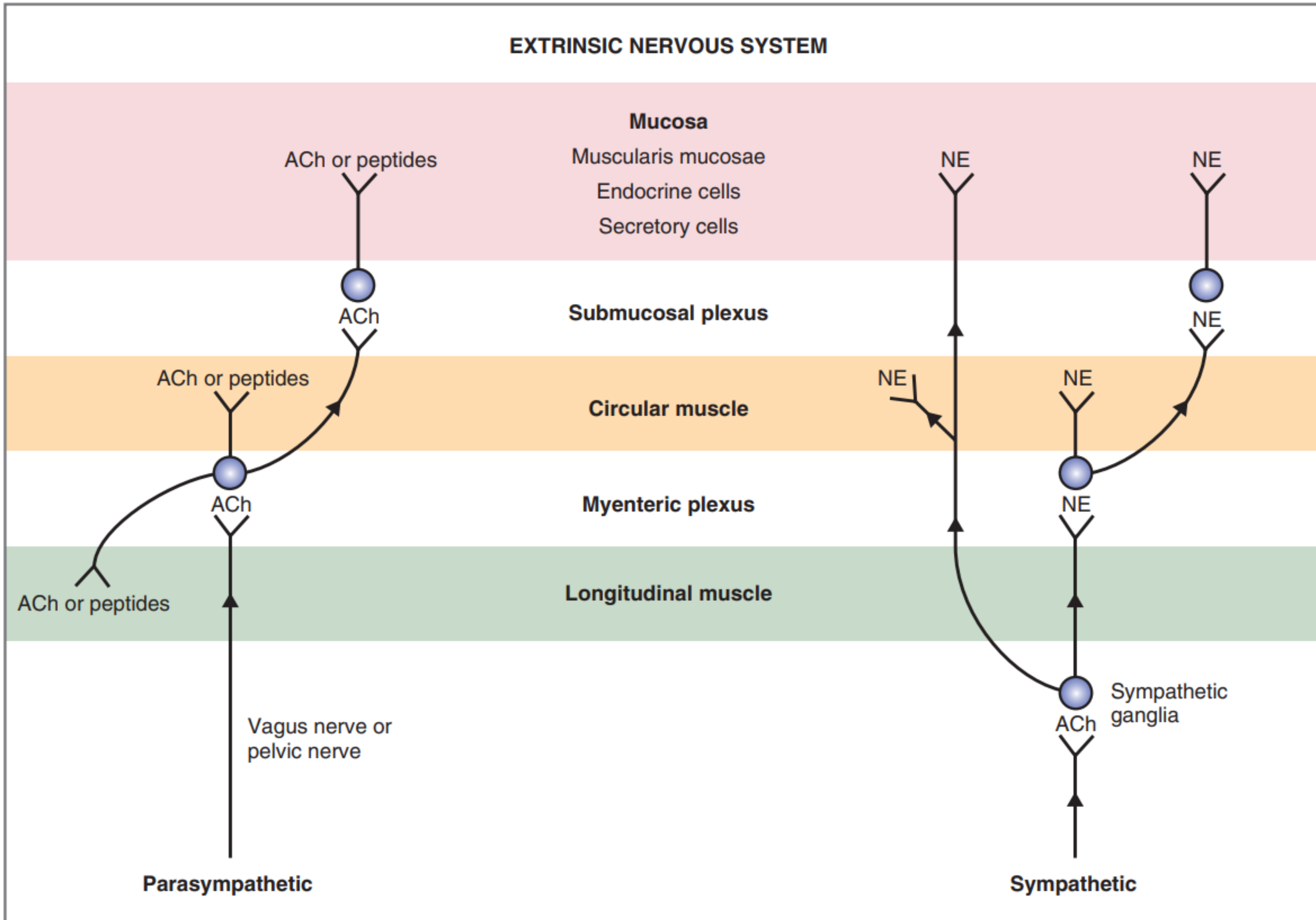


Fig. 8.2 The extrinsic nervous system of the gastrointestinal tract. Efferent neurons of the parasympathetic and sympathetic nervous systems synapse in the myenteric and submucosal plexuses, in the smooth muscle, and in the mucosa. *ACh,* Acetylcholine; *NE,* norepinephrine.

Intrinisic Neural Control

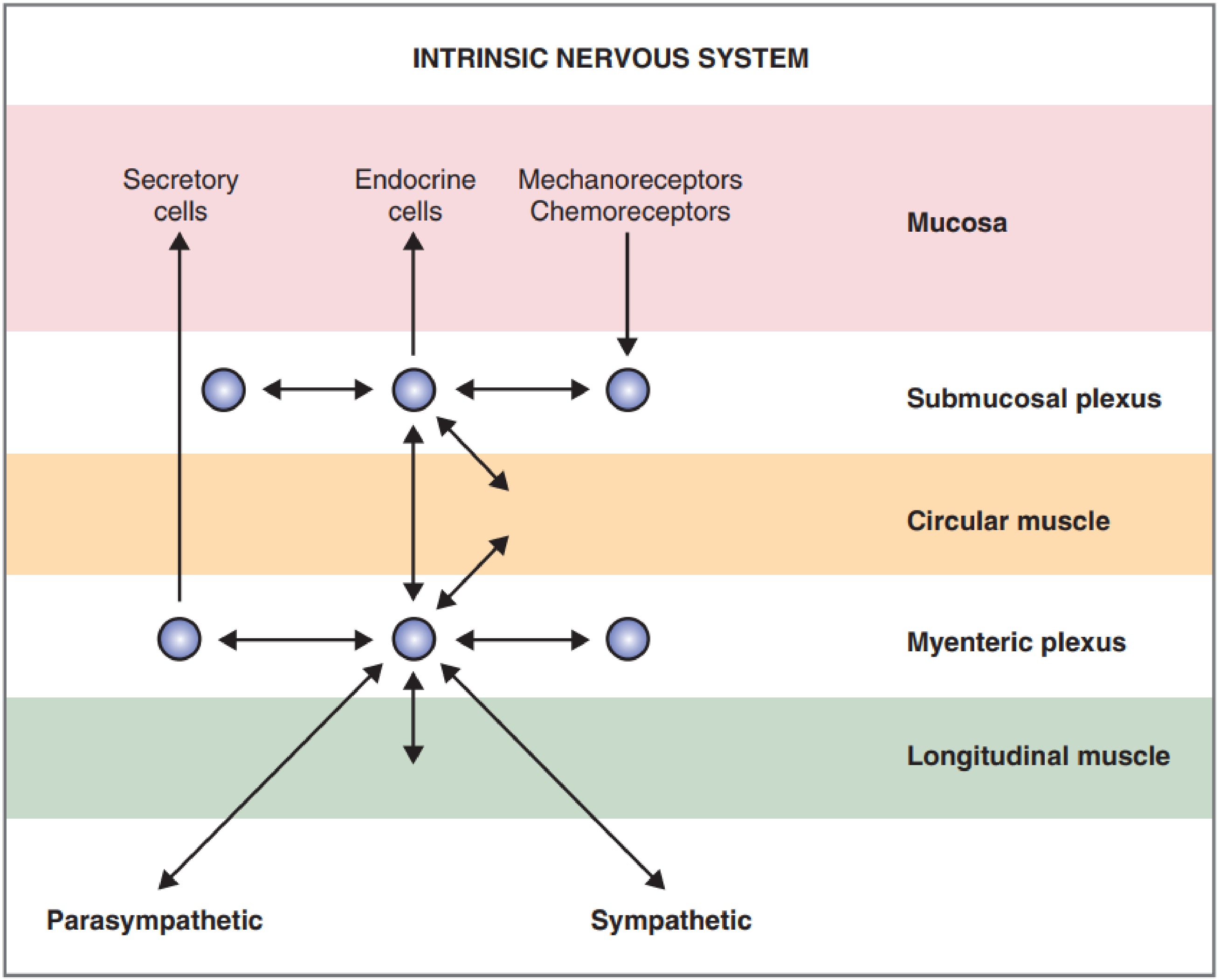


Fig. 8.3 Intrinsic nervous system of the gastrointestinal tract.

Clinical Correlate: Hirschsprung's disease

A 3-year-old boy is brought to his general practitioner again by his worried mother. She is concerned that he remains constipated despite trying a third different laxative. Further history showed that he passed his first meconium only on day 5, and since then has been opening his bowels only weekly, with associated straining. His growth has fallen from the 25th to the 2nd centile for height and weight. On examination he has a distended abdomen with palpable stool throughout the abdomen.

- **Aganglionic portion of large bowel**
- **Absent submucosal and myenteric plexi**
- **Massive distension of proximal colon**

What is Hirschsprung's disease?
Hirschsprung's disease is characterized by an absence of ganglion cells in the distal bowel, beginning at the internal sphincter and extending proximally. The resulting aganglionic segment of the colon fails to relax, causing a functional obstruction. Presentation is commonly in the first 28 days of life (neonatal period), with delayed passage of meconium and abdominal distension (see "Red flags" box). However, about 12% of patients present again in childhood with intractable constipation (not responsive to laxatives) and failure to thrive, with about a third of these presenting with enterocolitis.[1] [2]

"Red flags" for Hirschsprung's disease*
Delayed (>24 h) meconium—Present in 70-87% of cases of Hirschsprung's disease and in <1% of normal children[2] [3]
Neonatal constipation—Present in 90-95% of cases but in <7% of children with functional constipation[2] [3]
Family history (affected sibling)—Present in 12-33% of cases[4] [5]
Poor growth—Present in 25-30% of cases[2] [3]
Abdominal distension—Present in 76-85% of cases but in 20% of patients with functional constipation[2] [3]
Down's syndrome and other chromosomal anomalies—Hirschsprung's disease is present in 1.5% of patients with Down's syndrome, but 5-10% of patients with Down's have functional constipation[7]
*Three or more red flags are present in 18% of patients with the disease. No red flags are present in <1% of patients with the disease[2] [3]

Citation
BMJ 2012;345:e5521; https://www.bmj.com/content/345/bmj.e5521

GIT Control Summary

Regulation of GIT function is integrated via several coordinated responses:

1. Neural pathways
2. Peptides and other regulatory mediators

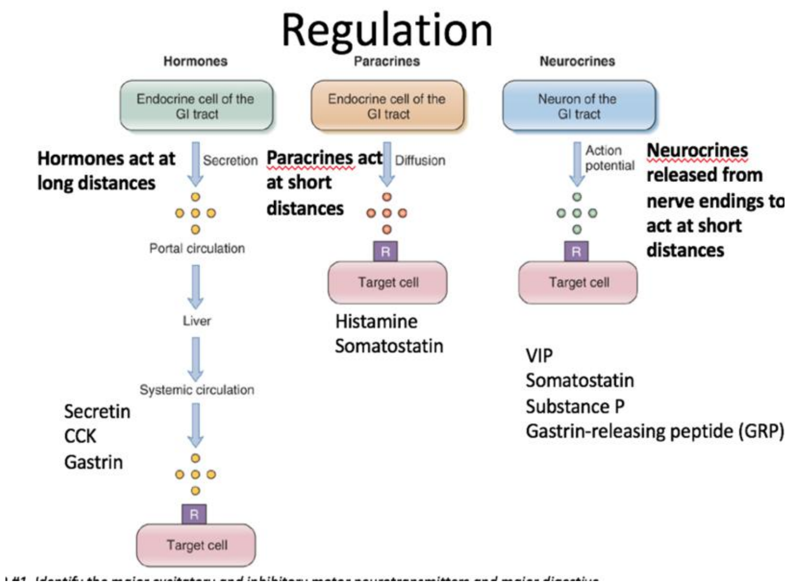
Summary of neural pathways:

- Short reflexes integrated in ENS (Enteric Nervous System)
- Long reflexes integrated in CNS (Central nervous system)
- Connections between CNS\ENS via ANS (Autonomic Nervous System)

Part 2

GIT Regulators

1. Endocrine/hormonal
2. Neurocrine
3. Paracrine



Neural, Hormonal and Paracrine Regulation

Neurocrine:

- Peptides released by the nerves within the gut
- Short-range effects after release
- ACh, Substance P (smooth muscle contraction),
- Norepinephrine, Vasoactive Intestinal Peptide (relaxation)

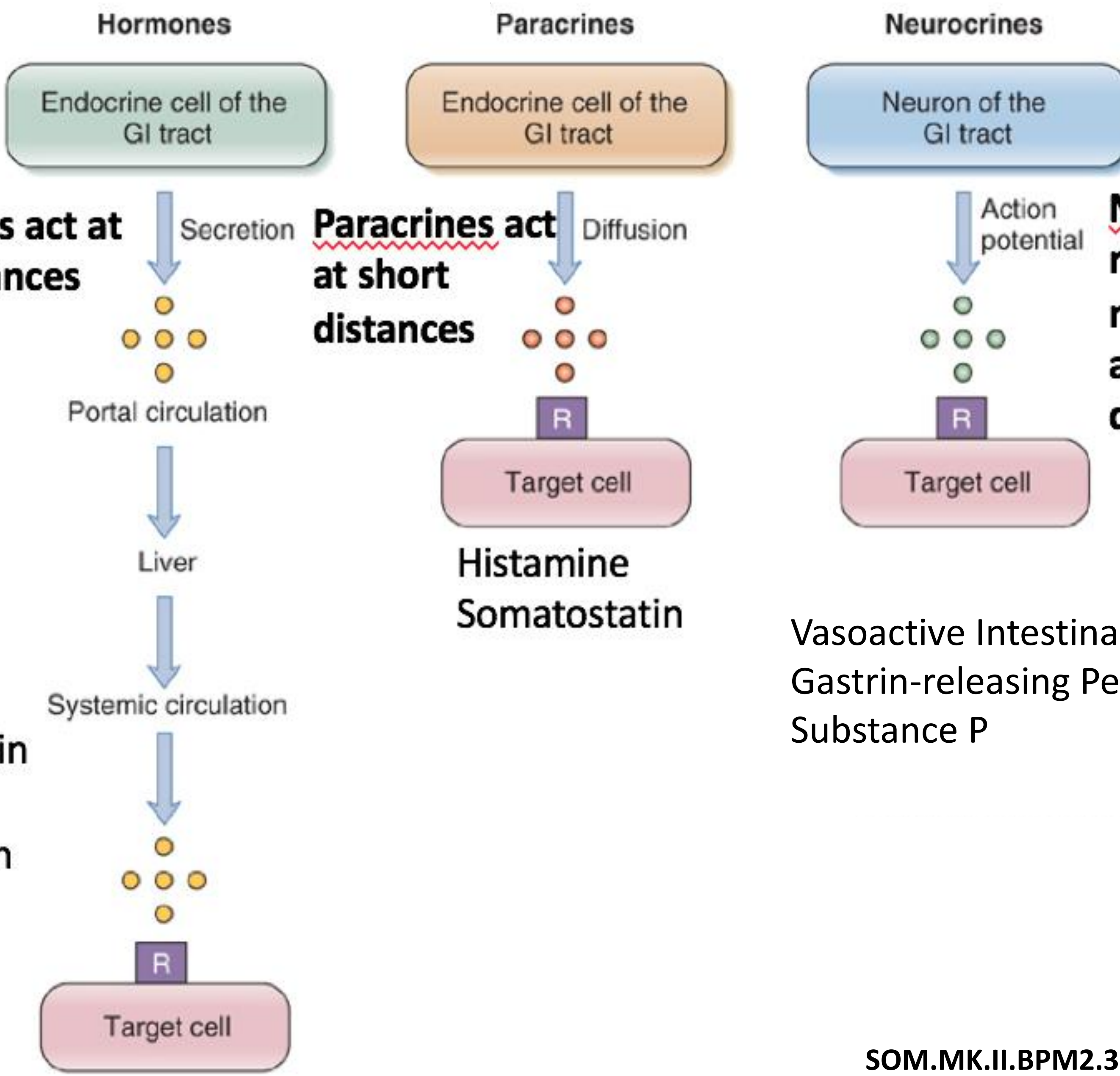
Endocrine/Hormonal:

- Hormones synthesized by cells in the gastric and intestinal epithelium and released into the blood stream
- Systemic long-range action
- Gastrin, secretin, CCK, the incretins (GIP & GLP)

Paracrine:

- Substances that act locally within the same tissue
- Short-range effects
- Histamine, somatostatin

GIT Regulators



Hormones act at long distances

Secretin
CCK
Gastrin

GIP
GLP

Paracrines act at short distances

Histamine
Somatostatin

Neurocrines released from nerve endings to act at short distances

Vasoactive Intestinal Peptide (VIP)
Gastrin-releasing Peptide (GRP)
Substance P

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Endocrine Regulators: Gastrin

- **Key role:** stomach acid regulation
- **Source**: G cells gastric antrum after a meal
- **Stimuli for release:**
  - Small peptides and amino acids in stomach
  - Stomach distension
  - Vagal stimulation, via by gastrin-releasing peptide (GrP)
- **Inhibition of release:**
  - high stomach H^+ (acidity)
  - Somatostatin
- **Effects:**
  - Increases H^+ secretion by the gastric parietal cells
  - Stimulates growth of gastric mucosa
  - Increases gastric motility.

Endocrine Regulators: Cholecystokinin (CCK)

- **Key role:** control of gastric emptying and pancreatic enzyme release
- **Source**: I cells in duodenum and jejunum
- **Stimuli for release**:
  - Small peptides, amino acids, fatty acids, monoglycerides
- **Inhibition of release**:
  - Absence of stimuli (via absorption)
  - Somatostatin
- **Effects**:
  - Stimulates pancreatic enzyme and HCO_3^- secretion
  - Slows gastric emptying to allow more time for intestinal digestion and absorption
  - Gallbladder contraction -> bile secretion (and pain if gallstones are present)
  - Stimulates hepatic bile production

Endocrine Regulators: Secretin

- **Key role:** Control of pancreatic HCO_3^- secretion
- **Source:** S cells in duodenum
- **Stimuli for release**
  - H^+ in the lumen of the duodenum
  - Fatty acids and peptides in the duodenal lumen
- **Inhibitors**
  - Neutral pH in duodenum
  - Somatostatin
- **Effects**
  - Stimulates pancreatic HCO_3^- and H_2O secretion -> neutralizes small intestine acidity from stomach acid
  - Inhibits H^+ secretion by gastric parietal cells

Incretins: GIP and GLP (endocrine)

- **Key role:** glucose & metabolism regulation, satiety, pacing
- **GIP** - Gastric Inhibitory peptide (also known as Glucose dependent insulinotropic peptide)
  - Source: K cells in duodenum and proximal jejunum
  - Stimuli: luminal glucose and fatty acids
- **GLP/GLP-1** - Glucagon like peptide
  - Source: L-cells in proximal jejenum – colon
  - Stimuli: mainly fatty acids and bile in ileum (other nutrients are already mostly absorbed)
  - Agonist drugs: Ozempic (type 2 DM) and Wegovy (weight loss)
- **Effects**
  - Increase insulin secretion from pancreas (both GIP & GLP)
  - Stimulates glucagon secretion from pancreas (GIP only)
  - Decreases glucagon secretion from pancreas (GLP only)
  - Decreased gastric H^+ secretion (GIP only)
  - Slowed gastric emptying (GLP only)
  - Satiety (GLP only)

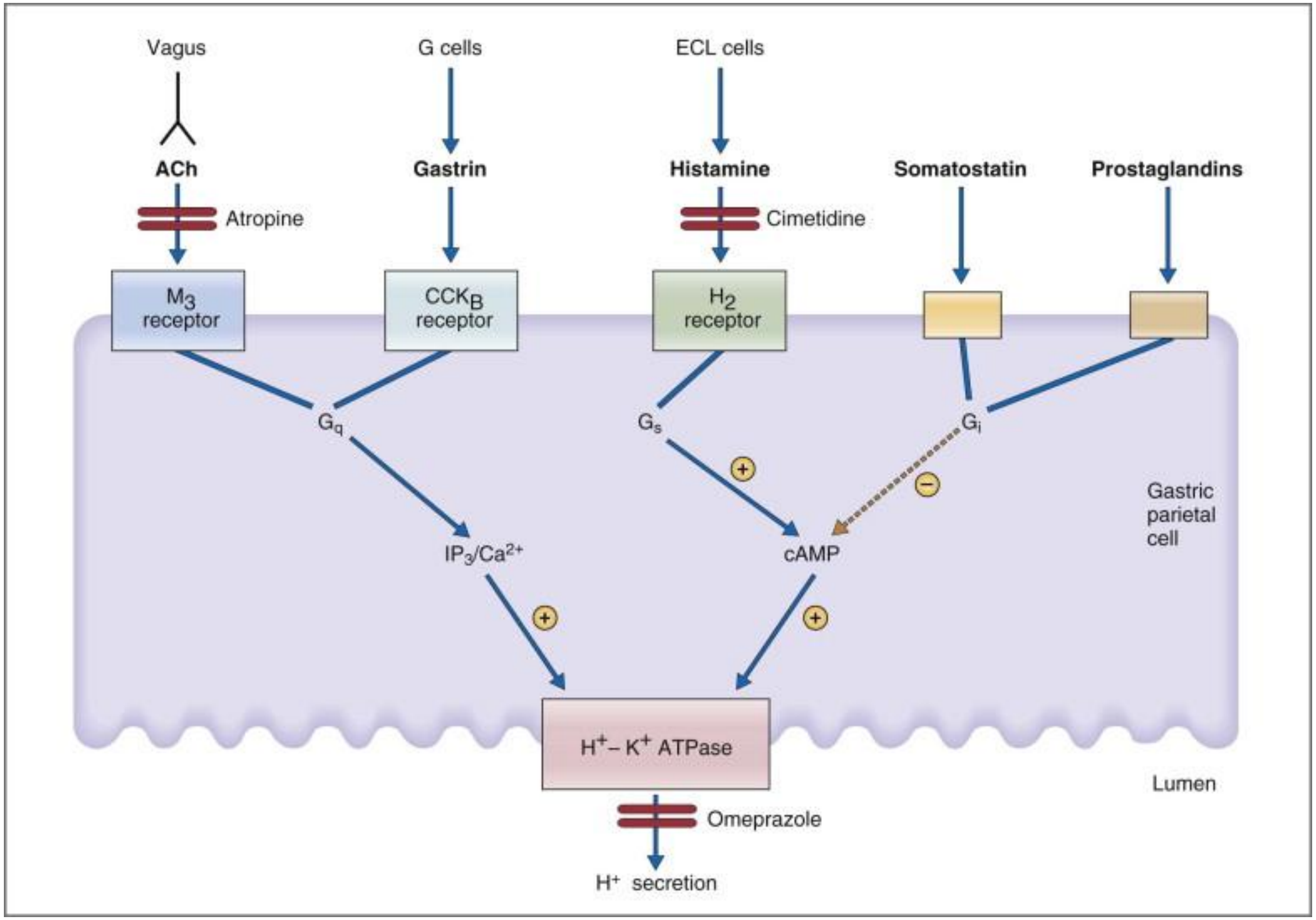
Paracrine Regulatory Mediators

- Released from endocrine cells in the GI mucosa
- Diffuse short distances to target cells in GIT
- **Somatostatin**
  - secreted throughout GIT in response to luminal H^+
  - secretion is inhibited by vagal stimulation
  - inhibits the release of all GI hormones
  - inhibits gastric H+ secretion
- **Histamine**
  - secreted by mast cells of the gastric mucosa
  - increases gastric H^+ secretion directly, and by potentiating the effects of gastrin and vagal stimulation

Select Neurocrine Regulators

- **Vasoactive Intestinal Peptide (VIP)**
  - homologous and functionally similar to secretin (limit acidification)
  - released by neurons in GIT mucosa and smooth muscle
  - relaxation of GI smooth muscle
  - stimulates pancreatic HCO_3^- secretion
  - inhibits gastric H^+ secretion
- **Nitric Oxide (NO)**
  - Released by ENS neurons throughout GI tract
  - Smooth muscle relaxation
- **Gastrin-releasing peptide (GRP, also bombesin)**
  - released from vagus nerve, stimulates gastrin release from G cells
- **Enkephalins (met-enkephalin and leu-enkephalin, opiates)**
  - secreted from ENS neurons in mucosa and smooth muscle
  - contraction of GI smooth muscle, particularly sphincters
  - inhibit intestinal fluid and electrolyte secretion
  - used in diarrhea treatment

Gastric parietal cell signal integration



End organ responses involve integration and overlap of signalling modes – neural (vagus nerve), hormone (gastrin) and paracrine (histamine, SST, prostaglandins).

Gastric parietal cell signal integration

Parietal cells are stimulated by neural (acetylcholine), endocrine (gastrin), and paracrine (histamine) mechanisms.

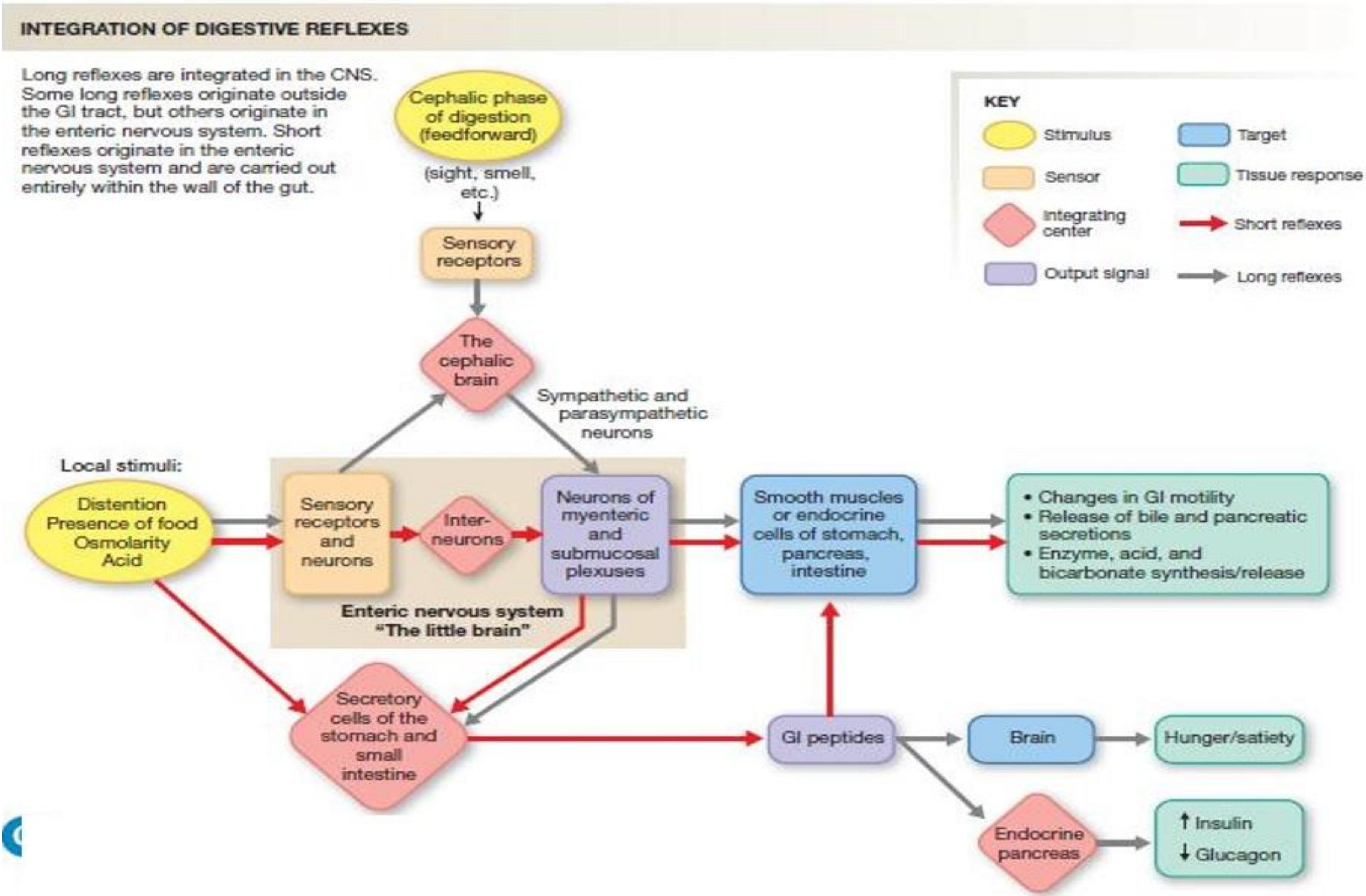
Neural mechanism: depolarization of vagal postganglionic nerve fibers releases ACh onto the muscarinic M_3 receptor on parietal cells. The activated M_3 receptor then upregulates the phospholipase C, IP_3, and diacylglycerol (DAG) signal transduction pathway, which targets the apical H^+/K^+-ATPase.

Paracrine mechanism: enterochromaffin cells release histamine onto H_2 receptors on parietal cells. Binding of histamine to the H_2 receptor activates the adenylate cyclase, cyclic adenosine monophosphate (cAMP), protein kinase A signal transduction pathway, leading to activation of the final common pathway, the apical H^+,K^+-ATPase.

Hormonal mechanism: gastrin is released from antral G cells in response to Ach (indirect effect of Ach), gastrin-releasing peptide, and amino acids. Gastrin circulates as a GI hormone, chiefly binding gastrin (or CCK_B) receptors on gastric parietal cells. Like ACh, gastrin stimulates parietal cell H^+ secretion through the $IP_3/DAG/Ca^{2+}$ second messenger system.

Inhibition of parietal cell H^+ secretion is mediated primarily via the paracrine effects of somatostatin, prostaglandins, and adenosine. All three paracrines inhibit parietal cell adenylate cyclase and cAMP production. Somatostatin has the additional effect of both inhibiting histamine release from enterochromaffin cells and gastrin release from antral G cells.

Integration of digestive reflexes



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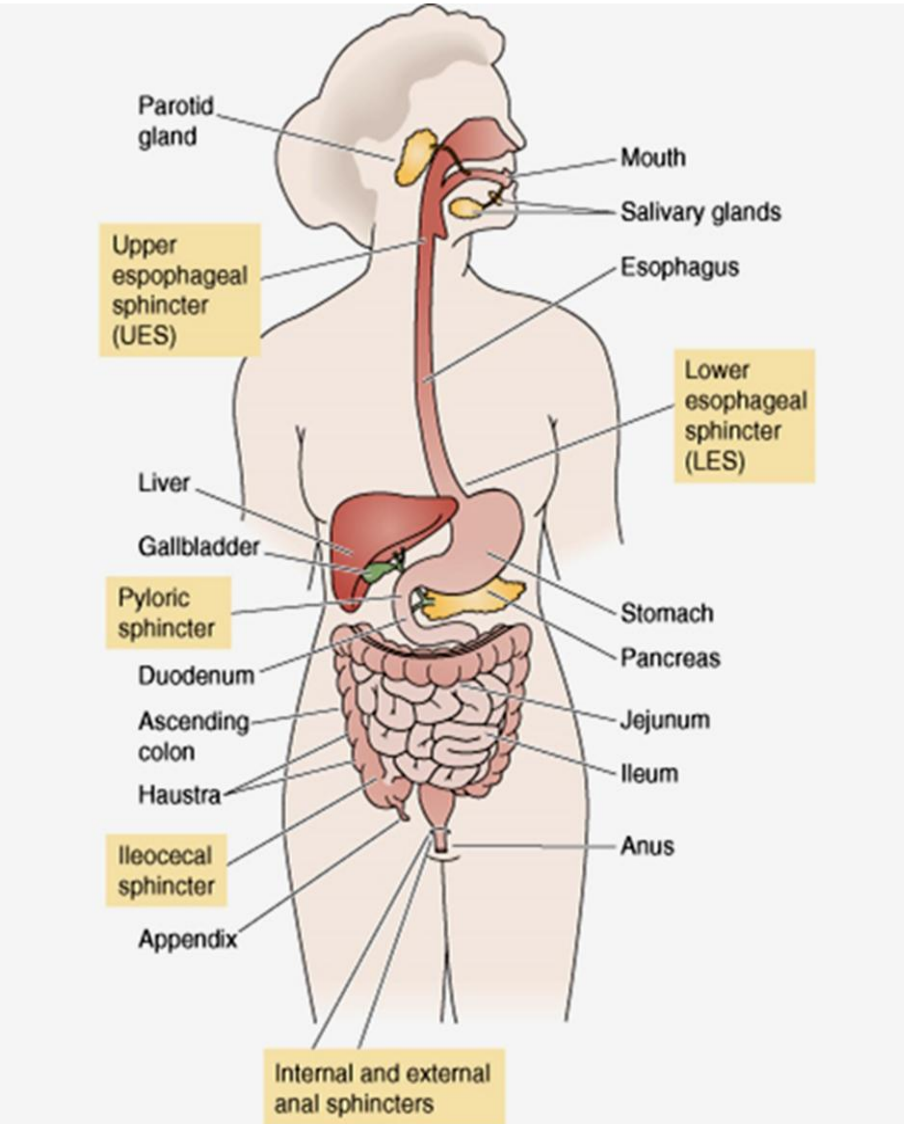
Part 3

Stages of Meal Processing

3 digestive phases after a meal



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1. Cephalic

Anticipation, sight, smell and taste of food

2. Gastric

Food is in stomach

3. Intestinal

Food is intestines

Cephalic Phase

Food "thoughts", smells, sight →

↑ salivary secretion

↑ gastric, pancreatic and intestinal secretions

↑ motility

<u>Mediated by long reflexes</u>

Afferents: sight, smell, food in mouth

Coordinating centre: Medulla oblongata

Effectors: Vagus nerve, ENS

Gastric Phase

Stimulus: Stomach distension and the presence of peptides and amino acids in stomach.

Coordinating Centre: Local endocrine cells and ENS.

Effectors: Hormones, paracrines and neurotransmitters.

Actions: ↑ secretions and motility.

Results:

- digestion of proteins in stomach by pepsin
- formation of chyme
- controlled entry of chyme into the SI

Intestinal Phase

Trigger: Entry of chyme into the SI

Actions: A series of reflexes that aid the entry of chyme into SI to promote digestion and motility.

Central Appetite Regulation

- **Leptin** ('satiety hormone') is secreted by fat cells. It stimulates neurons that decrease appetite and inhibit those that increase appetite.
- **Insulin and GLP-1** also inhibit appetite
- **Ghrelin** ('hunger hormone') is secreted by gastric cells. It has the opposite of leptin, i.e. increasing appetite.

All of the above, as well as blood glucose levels, act on feeding centers in the hypothalamus, which integrates the signals into an overall hunger/satiety feeling. More detail in ANS lectures in NB module.

Next Steps

- Map out the GIT and list the functions of each part
- Make an overview of the key GI regulators and their function, sources and stimuli for release/termination
- Use the subsequent self-study slides for revision
- **Next lecture: Motility**

The following slides are for self- study and include summary slides as well as revision questions.

Please use these as one of your revision resources.

Summary: What are the key functions of the GIT?

Digestion	Mechanical and Chemical breakdown	- Mastication - Digestive juices
Endocrine	Hormones released into general circulation and other parts of the GIT	- Endocrine - Paracrine
Motility	Propulsion moves chyme along the tract	- Loss of motility results in dysfunction
Absorption	Nutrients, electrolytes, water, vitamins, etc.	- Villi of the enterocytes of the SI
Storage	Accommodation of food in stomach and large intestine	- Receptive relaxation

Self study: Can you think of additional functions?

Expulsion	Excretion of fecal waste matter via defecation	• Non absorbed food, colonic bacteria, substances e.g. drugs
Protection	Degradation of bacteria or antigens or other harmful substances	• Immune activity – Gut Associated Lymphoid Tissue (GALT), Peyer patches, lymphocytes • Immunoglobulin A • Gastric acid
Secretion (other)	Mucus, buffers, hormones, and enzymes facilitates lubrication, digestion, motility and absorption	• GIT secretes 7-9 L of fluid per day

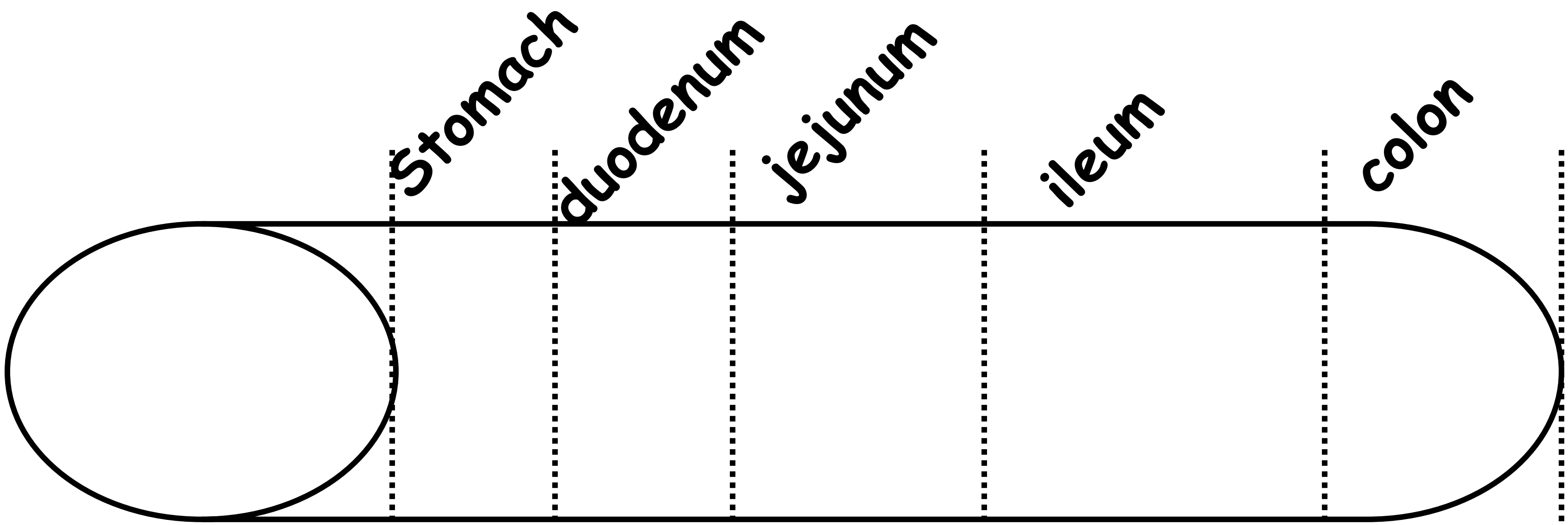
Self Study: Summary of GIT Structure/Function

TRACT	
Oral cavity	Digestion begins here Chewing breaks up food particles Moves food in the oral cavity Saliva secretion – Lubrication, Protection, and Digestion Lubrication with mucous which moistens particles, easier to swallow Protection from effects of oral bacteria. Amylase a carbohydrate-digesting enzyme and lingual lipase (fats)
Pharynx and Esophagus	No digestion occurs Provide a pathway for propulsion of the bolus from mouth to stomach.
Stomach	Stores food, Breaks food up into small particles and Partially Digests it. Secretes HCL and other exocrine secretions. Delivers fluid and partially digested food in small amounts to the small intestine. N.B. – Absorption of a few substances e.g. alcohol and some drugs. Little direct effect on fats/carbs.

Self Study: Summary of GIT Structure/Function

Small Intestines	Final stages of digestion and most absorption occurs here. Carbohydrates to **Monosaccharides** Fat to fatty acids Protein to amino acids. Vitamins, minerals and water also absorbed
Large Intestine	Store undigested material Acted on by bacteria Some water, electrolytes and carbohydrate reabsorption Defecation when portion nearest anal sphincter is distended.
GLANDS	
Pancreas	Releases enzymes into the duodenum Releases HCO_3^- to neutralize HCl.
Liver	Bile salts production: helps to breakdown fat (emulsification)

Self-study: Where are some of the major gut peptides secreted?



gastrin

cholecystokinin

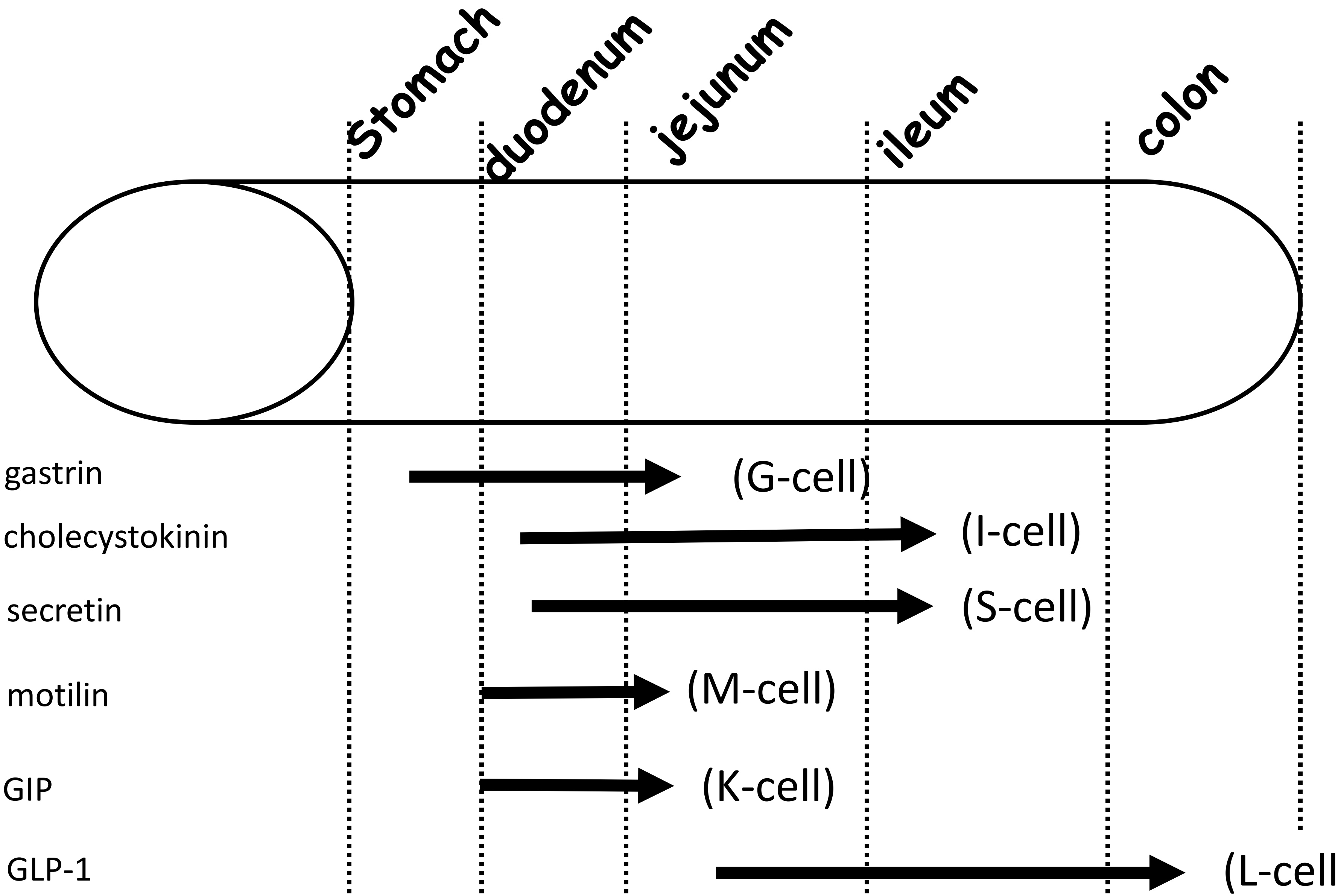
secretin

motilin

GIP

GLP-1

Where are some of the major gut peptides secreted? Answer.



Summary: Endocrine Regulatory Peptides

Peptide	(+) secretion	(-) secretion	Site of Secretion	Target Cells	Action
Gastrin	AA in stomach, stomach distension, Gastrin releasing peptide, Ach (Vagus)	Acid pH<1.5, Somatostatin	**G** cells of antrum, duodenum, jejunum	Entero-chromaffin like cell(ECL) Parietal cells	Stimulates gastric acid ↑ gastric mucosal growth ↑Gastric motility
CCK	FA, AA, distension	Somatostatin	**I** cells of duodenum jeunum and ileum	Secretion in circulation	*Motility of gall bladder *reduced delivery of chyme from stomach *↑pancreatic secretions *↑satiety
Secretin	Acid in SI	Somatostatin	S cells of duodenum, jejunum and ileum	Pancreas Stomach	↓gastric acid ↑HCO3 secretion by pancreas Neutralizes acid delivered to duodenum Optimum pH for intestinal enzymes
Glucose dependent insulin-tropic polypeptide GIP ***GLP-1**	FA, AA, CHO	Neuropeptides	K cells of duodenum and jejunum Upper GI	β Cells of pancreas	Stimulates insulin release Inhibits gastric acid secretion

Summary: Neurocrine Regulatory Peptides

Peptide	(+) secretion	(-) secretion	Site of Secretion	Target Cells	Action
Vasoactive Intestinal peptide	Released by action potential	Somatostatin	Neurons in the mucosa and smooth muscle of the GIT	Smooth Muscles including Lower esophageal sphincter	• Relaxation of the GI smooth muscle • Pancreatic HCO_3^- secretion • Inhibits gastric H^+ secretion
Gastrin Releasing Peptide	Vagus nerve	Acid in the stomach	Post ganglionic fibers of Vagus nerve that innervate the G cells	G cells	*Stimulates gastrin secretion

Summary: Paracrine Regulatory Peptides

Peptides	**(+) secretion**	**(-) secretion**	**Site of Secretion**	**Target Cells**	**Action**
Somatostatin	H+ in the lumen	Vagal stimulation	**D cells of stomach** (also from pancreas D cells, but paracrine only)	Potent Inhibitor	➢(↓) pancreatic /gastric secretion ➢(↓)motility ➢(↓)gall bladder contraction ➢(↓)nutrient absorption ➢vasoconstrictor
Histamine	Food in the stomach	H+ in stomach	Entero-chromaffin like cell ECL, mast cells	Specific receptors	*Stimulates gastric acid secretion Enhances vascular vasodilation

Important GIT neurotransmitters

Substance	Source	Actions
Acetylcholine (ACh)	Cholinergic neurons	Contraction of smooth muscle in wall Relaxation of sphincters ↑ Salivary secretion ↑ Gastric secretion ↑ Pancreatic secretion
Norepinephrine (NE)	Adrenergic neurons	Relaxation of smooth muscle in wall Contraction of sphincters ↑ Salivary secretion
Vasoactive Intestinal Peptide (VIP)	Neurons of the enteric nervous system	Relaxation of smooth muscle ↑ Intestinal secretion ↑ Pancreatic secretion
Nitric Oxide (NO)	Neurons of the enteric nervous system	Relaxation of smooth muscle
Gastrin-Releasing Peptide (GRP), or Bombesin	Vagal neurons of gastric mucosa	↑ Gastrin secretion
Enkephalins (Opiates)	Neurons of the enteric nervous system	Contraction of smooth muscle ↓ Intestinal secretion
Neuropeptide Y	Neurons of the enteric nervous system	Relaxation of smooth muscle ↓ Intestinal secretion
Substance P	Cosecreted with ACh by neurons of the enteric nervous system	Contraction of smooth muscle ↑ Salivary secretion